

Requirements in System Design

After you finish your base SVD, i.e. the last PDF, you want to *create something* called a **REQUIREMENTS DOCUMENT**.

Simply, what should the application we're making do? And this is split into a couple categories.

> **Functional:** What the system can actually do, i.e. check out books, return books, scan codes; basically 1.3 from SVD, *but in more detail*

> **Usability:** How user friendly and easy to use will it be? (UX/UI)

> **Reliability:** How long can the client expect your system to be up and running? (99.9%? 99.99%? 99.999%) *those decimals matter 9hrs, 52mins 5mins.*

> **Performance:** How well does it run, i.e. how quickly can it do a search?

> **Security:** What security practices are on the system, i.e. don't give every user full access to view data.

> **Implementation:** At a BASE level, what technical tools are you going to use, like Python, HTTPS, SQL (*tech stack*)

> **Interface:** What other systems or tools can our system connect to? Maybe an external payment processor.

> **Physical:** Assume our application isn't just software, but also hardware, *what dimensions and weight would it need?*

> **Support:** How will you as the developer help the client with issues they're facing? (24/7? 9-5pm?)

You set standards relating to these topics, let's do one for the *library system from the last part*.

Functional

- Check out and return books via barcode scan
- Search catalog, calculate fines, track patrons

Usability

- Checkout takes under 2 minutes for new staff (Ease Of Use)
- Common tasks within 2 clicks (Ease Of Use)

Reliability

- 99.9% uptime (8.7 hours downtime/year)
- Available during library hours

Performance

- Search results under 3 seconds
- Supports 10 concurrent users

Security

- Staff login with roles (librarian, manager)
- Patrons see only their own records

Implementation

- Python + SQLite + REST API
- Runs on library computers

Interface

- USB barcode scanners and receipt printers
- Email server for overdue notices

Physical

- Standard Windows desktop computers
- Handheld barcode scanners

Support

- 9am-5pm email support
- 4-hour response for critical issues